

Ahlfors 4

Vincent Wang

May 2026

1 Fundamental Theorems

$f = u(t) + iv(t)$ is continuous for $t \in [a, b]$. Define

$$\int_a^b f(t)dt = \int_a^b u(t)dt + i \int_a^b v(t)dt.$$

If $c = \alpha + i\beta \in \mathbb{C}$, then

$$\begin{aligned} \int_a^b (u(t) + iv(t))(\alpha + \beta i)dt &= \int_a^b (\alpha u(t) - \beta v(t)) + (\beta u(t) + \alpha v(t))idt \\ &= \int_a^b (\alpha u(t) - \beta v(t))dt + i \int_a^b (\beta u(t) + \alpha v(t))dt = \alpha \int_a^b u(t)dt - \beta \int_a^b v(t)dt + i\beta \int_a^b u(t)dt + i\alpha \int_a^b v(t)dt \\ &= (\alpha + \beta i) \left(\int_a^b u(t)dt + i \int_a^b v(t)dt \right) = (\alpha + \beta i) \int_a^b f(t)dt. \end{aligned}$$

Claim: $\left| \int_a^b f(t)dt \right| \leq \int_a^b |f(t)|dt$.

Proof: Let $\theta \in \mathbb{R}$. We have

$$\Re e^{-i\theta} \int_a^b f(t)dt = \Re \int_a^b e^{-i\theta} f(t)dt \leq \int_a^b |f(t)|dt.$$

Take $\theta = \arg \int_a^b f(t)dt$ (if $\int_a^b f(t)dt = 0$, there is nothing to show).

$$\int_{\gamma} f(z)d\bar{z} = \int_a^b f(\gamma(t))\overline{\gamma'(t)}dt = \int_a^b \overline{f(\gamma(t))\gamma'(t)}dt = \overline{\int_{\gamma} f dz}.$$

Define

$$\begin{aligned} \int_{\gamma} f dx &= \frac{1}{2} \left(\int_{\gamma} f dz + \int_{\gamma} f d\bar{z} \right) \\ \int_{\gamma} f dy &= \frac{1}{2i} \left(\int_{\gamma} f dz - \int_{\gamma} f d\bar{z} \right). \end{aligned}$$

$$\left| \int_{\gamma} f dz \right| = \left| \int_a^b f \gamma(t) \gamma'(t) dt \right| \leq \int_a^b |f \gamma(t)| \cdot |\gamma'(t)| dt = \int_{\gamma} |f| \cdot |dz|$$

Suppose γ given by $z : I \rightarrow \mathbb{C}$ is differentiable on $(0, 1)$ and continuous on $[0, 1]$. Let $z(t) = u(t) + iv(t)$, so u, v are diff on $(0, 1)$ and cont on $[0, 1]$. Let $0 = t_0 < t_1 < t_2 < \dots < t_n = 1$ be a partition of I . By MVT, $u(t_{k+1}) - u(t_k) = u'(s_k)(t_{k+1} - t_k)$ for some $t_k < s_k < t_{k+1}$. We have a similar formula for v . Let r_k denote the values in between.

Then

$$\sum_{k=0}^{n-1} |z_{k+1} - z_k| = \sum_{k=0}^{n-1} (t_{k+1} - t_k) \sqrt{u'(s_k)^2 + v'(r_k)^2}$$

Claim: Let $A \neq \emptyset$ be such that for all $\epsilon > 0$, for all $x \in A$, there exists $y \geq x$ in A such that $|y - \alpha| < \epsilon$. Then $\sup A = \alpha$.

Proof: Let $\epsilon = 1$ and $x \in A$. Then $x < y < \alpha + 1$, so A is bounded above by $\alpha + 1$, so $\sup A$ exists. Let $\epsilon > 0$ let $x \in A$ be arbitrary. Then there exists $y > x$ such that $|y - \alpha| < \epsilon$. In particular, $y > \alpha - \epsilon$, so $\sup A \geq y > \alpha - \epsilon$. Since this holds for all ϵ , we have $\sup A \geq \alpha$. Conversely, $x < y < \alpha + \epsilon$, so $\sup A \leq \alpha + \epsilon$. Since this holds for all ϵ , we have $\sup A \leq \alpha$. The result follows.

Note $|z(t_2) - z(t_0)| \leq |z(t_2) - z(t_1)| + |z(t_1) - z(t_0)|$. Hence, the differences $t_{k+1} - t_k$ can be taken to be arbitrarily small. Let $\epsilon > 0$. Assume u, v are continuously differentiable. Then u, v are uniformly continuous on I . Pick $\delta > 0$ such that $|t - t'| < \delta \rightarrow |u(t) - u(t')|, |v(t) - v(t')| < \epsilon$. Pick the t_k such that $t_{k+1} - t_k < \delta$ for all k .

Then $(u(t_k) - u(s_k))^2 \leq \epsilon^2$ and $(v(t_k) - v(r_k))^2 \leq \epsilon^2$, so

$$\begin{aligned} \sum_{k=0}^{n-1} |z_{k+1} - z_k| &\leq \sum_{k=0}^{n-1} (t_{k+1} - t_k) \left(\sqrt{u'(t_k)^2 + v'(t_k)^2} + \sqrt{(u'(t_k) - u'(s_k))^2 + (v'(t_k) - v'(r_k))^2} \right) \\ &\leq \sum_{k=0}^{n-1} (t_{k+1} - t_k) \left(\sqrt{u'(t_k)^2 + v'(t_k)^2} + 2\epsilon \right) \\ &= \sum_{k=0}^{n-1} (t_{k+1} - t_k) \sqrt{u'(t_k)^2 + v'(t_k)^2} + 2\epsilon \\ &\leq \int_a^b \sqrt{u'(t)^2 + v'(t)^2} dt + 2\epsilon = \int_{\gamma} |dz| + 2\epsilon. \end{aligned}$$

(can be rewritten to consider the difference, so we apply the claim above).

2 Review

Suppose $f(z) = \sum_{n=0}^{\infty} a_n z^n$ has radius of convergence $R > 0$. Let $z_0 \in \mathbb{C}$ such that $|z_0| = R$ and $f(z_0)$ converges. Then $\lim_{z \rightarrow z_0; |z_0 - z|/(R - |z|) \leq M} f(z) = f(z_0)$.

Proof: Let $b_n = a_n z_0^n$. Define $g(w) = \sum_{n=0}^{\infty} b_n w^n$. Note that g has radius of convergence 1 and $g(1) = f(z_0)$. It suffices to show

$$\lim_{w \rightarrow 1; |1-w|/(1-|w|) \leq M} g(w) = g(1).$$

Let $h(w) = g(w) - g(1) = \sum_{n=0}^{\infty} c_n w^n$, so $\sum_{n=0}^{\infty} c_n = 0$. It suffices to show

$$\lim_{w \rightarrow 1; |1-w|/(1-|w|) \leq M} h(w) = 0.$$

Define $s_n = \sum_{k=0}^n c_k$. We have

$$\begin{aligned} h(z) &= \lim_{N \rightarrow \infty} \left(c_0 + \sum_{n=1}^N (s_n - s_{n-1}) z^n \right) = \lim_{N \rightarrow \infty} \left(c_0 + \sum_{n=1}^N s_n z^n - z \sum_{n=1}^N s_{n-1} z^{n-1} \right) \\ &= \lim_{N \rightarrow \infty} \left(\sum_{n=0}^N s_n z^n - z \sum_{n=0}^{N-1} s_n z^n \right) = \lim_{N \rightarrow \infty} \left((1-z) \sum_{n=0}^{N-1} s_n z^n + s_N z^N \right) \\ &= (1-z) \sum_{n=0}^{\infty} s_n z^n. \end{aligned}$$

Let $\epsilon > 0$. Pick $N \in \mathbb{Z}^+$ such that $n \geq N \rightarrow |s_n| < \epsilon$. Let $C = \sum_{n=0}^{N-1} |s_n|$. Then

$$\begin{aligned} |h(w)| &= \left| (1-w) \left(\sum_{n=0}^{N-1} s_n w^n + \sum_{n=N}^{\infty} s_n w^n \right) \right| \\ &\leq |1-w| \cdot \sum_{n=0}^{N-1} |s_n| + |1-w| \cdot \sum_{n=N}^{\infty} |s_n| \cdot |w|^n \\ &\leq |1-w| \cdot C + |1-w| \cdot \epsilon / (1-|w|) \\ &\leq |1-w| \cdot C + M\epsilon. \end{aligned}$$

Pick $\delta = M\epsilon/(C+1)$. Then $|h(w)| \leq 2M\epsilon$. Since M is constant, the limit follows.

Let $\Omega, \Omega' \subseteq \mathbb{C}$ be regions. A function $f : \Omega \rightarrow \Omega'$ is said to be conformal at $z_0 \in \Omega_1$ if (i) $\lim_{z \rightarrow z_0} \frac{|f(z) - f(z_0)|}{|z - z_0|} \neq 0$ exists, and (ii) if $z, w \neq z_0$, then

$$\lim_{r \rightarrow 0; r > 0} \arg \frac{f((1-r)z_0 + rz) - f(z_0)}{f((1-r)z_0 + rw) - f(z_0)} = \arg \frac{z - z_0}{w - z_0}.$$

First, I claim there exists $\delta > 0$ such that $|z - z_0| < \delta \rightarrow f(z) \neq f(z_0)$. Suppose FTSOC for all $n \in \mathbb{Z}^+$, we can pick $0 < |z_n - z_0| < 1/n$ such that $f(z_n) = f(z_0)$. Then

$$\lim_{n \rightarrow \infty} \frac{|f(z_n) - f(z_0)|}{|z_n - z_0|} = 0,$$

absurd. Hence, if the first limit exists, then the second limit is reasonable.

First, suppose f is analytic such that $f'(z_0) \neq 0$. I claim that f is conformal at z_0 . Indeed,

$$\left| \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \right| = \lim_{z \rightarrow z_0} \left| \frac{f(z) - f(z_0)}{z - z_0} \right| = |f'(z_0)| \neq 0$$

and

$$\begin{aligned} & \lim_{r \rightarrow 0; r > 0} \arg \frac{f((1-r)z_0 + rz) - f(z_0)}{f((1-r)z_0 + rw) - f(z_0)} \\ &= \lim_{r \rightarrow 0; r > 0} \arg \frac{f((1-r)z_0 + rz) - f(z_0)}{r(z - z_0)} \cdot \frac{r(w - z_0)}{f((1-r)z_0 + rw) - f(z_0)} \cdot \frac{z - z_0}{w - z_0} \\ &= \arg \lim_{r \rightarrow 0; r > 0} f'(z_0)/f'(z_0) \cdot \frac{z - z_0}{w - z_0} = \arg \frac{z - z_0}{w - z_0}. \end{aligned}$$

Assume $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are continuous. Let $w = f(z(t))$. Then

$$\begin{aligned} w'(t) &= \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial t} \\ &= \frac{1}{2} \cdot \left[\frac{\partial f}{\partial x} \cdot (z'(t) + \overline{z'(t)}) - i \frac{\partial f}{\partial y} \cdot (z'(t) - i \overline{z'(t)}) \right] \\ &= \frac{1}{2} \cdot \left[\left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) \cdot z'(t) + \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \cdot \overline{z'(t)} \right] \\ \frac{w'(t)}{z'(t)} &= \frac{1}{2} \cdot \left[\left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) + \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \cdot \frac{\overline{z'(t)}}{z'(t)} \right] \end{aligned}$$

Suppose $w'(t)/z'(t)$ has constant argument. Then $\frac{\partial f}{\partial x} = -i \frac{\partial f}{\partial y}$, which is the Cauchy-Riemann equations. Suppose $w'(t)/z'(t)$ has constant magnitude. Then $\frac{\partial f}{\partial x} = i \frac{\partial f}{\partial y}$, so $\bar{f}$ is analytic, or f is analytic.

3 Back to Fundamental Theorems

Claim: The line integral $\int_{\gamma} p dx + q dy$ defined in Ω depends only on the end points of γ iff there exists a function $U(x, y)$ in Ω such that $\frac{\partial U}{\partial x} = p, \frac{\partial U}{\partial y} = q$.

Proof: Suppose the existence of U . Let γ be given by the differentiable path $z : I \rightarrow \Omega$. Let $z = x + yi$. Then

$$\begin{aligned} U(z_1) - U(z_0) &= \int_0^1 (Uz)' dt = \int_0^1 \left(\frac{\partial U}{\partial x} x'(t) + \frac{\partial U}{\partial y} y'(t) \right) dt \\ &= \int_{\gamma} p dx + q dy. \end{aligned}$$

Now suppose integrals are independent of paths. Fix $z_0 \in \Omega$. Define $U(z) = \int_{\gamma} p dx + q dy$, where γ is any path from z_0 to z . By assumption, U is well-defined. Now,

$$\begin{aligned} \frac{\partial U}{\partial x}(z) &= \lim_{t \rightarrow 0; t \neq 0} \frac{U(z+t) - U(z)}{t} \\ &= \lim_{t \rightarrow 0; t \neq 0} \frac{1}{t} \int_0^t p(z+s) ds = p(z) \end{aligned}$$

and similarly

$$\begin{aligned} \frac{\partial U}{\partial y}(z) &= \lim_{t \rightarrow 0; t \neq 0} \frac{U(z+it) - U(z)}{t} \\ &= \lim_{t \rightarrow 0; t \neq 0} \frac{1}{t} \int_0^t q(z+is) ds = q(z). \end{aligned}$$

Compute $\int_{\gamma} x dz$, where γ is the directed line segment from 0 to $1+i$. We have

$$I = \int_0^1 t^2(1+i) dt = \frac{1+i}{3}.$$

Compute $\int_{|z|=r} x dz$.

$$\begin{aligned} \int_{|z|=r} x dz &= \int_0^{2\pi} (r \cos \theta)(-r \sin \theta + ir \cos \theta) d\theta \\ &= ir^2 \int_0^{2\pi} \cos^2 \theta d\theta = \pi ir^2. \end{aligned}$$

$$\begin{aligned} \int_{|z|=r} x dz &= \frac{1}{2} \int_{|z|=r} (z + r^2/z) dz = \frac{r^2}{2} \int_{|z|=r} z^{-1} dz \\ &= \frac{r^2}{2} \int_0^{2\pi} r^{-1} e^{-i\theta} r e^{i\theta} i d\theta = i\pi r^2. \end{aligned}$$

Compute $\int_{|z|=1} |z-1| \cdot |dz|$.

$$\begin{aligned} \int_{|z|=1} |z-1| \cdot |dz| &= \int_0^{2\pi} \sqrt{2-2\cos \theta} d\theta = \int_0^{2\pi} 2|\sin(\theta/2)| d\theta \\ &= 2 \int_0^{2\pi} \sin(\theta/2) d\theta = 4 \int_0^{\pi} \sin \theta d\theta = 8. \end{aligned}$$

Suppose f is analytic on γ . Show that $\int_{\gamma} \overline{f(z)} f'(z) dz$ is purely imaginary.

Note that $(f(z))' = 2f(z)f'(z)$, so $\int_{\gamma} f(z)f'(z)dz = 0$ for all γ . It suffices to show $\int_{\gamma} \Re(f(z))f'(z)dz$ is purely imaginary. Let $f(x, y) = u(x, y) + iv(x, y)$. Then $f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$. Let $dz = dx + idy$, so we wish to show $\int_{\gamma} u \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \right) (dx + idy)$ is purely imaginary, or $\int_{\gamma} u \left(\frac{\partial u}{\partial x} dx - \frac{\partial v}{\partial x} dy \right) = 0$. Let $U(x, y) = \frac{1}{2}u(x, y)^2$. We have $\frac{\partial U}{\partial x} = u\frac{\partial u}{\partial x}$ and $\frac{\partial U}{\partial y} = u\frac{\partial u}{\partial y} = -u\frac{\partial v}{\partial y}$, so the differential $u \left(\frac{\partial u}{\partial x} dx - \frac{\partial v}{\partial x} dy \right)$ is exact and the result follows.

$\log$ is analytic on $\mathbb{C} - \mathbb{R}_{\leq 0}$, which contains the disc $|z-1| < 1$. Then $f'(z)/f(z) = (\log f(z))'$. $\log f(z)$ is analytic (as the composition of two analytic functions).

If $P(z)$ is a polynomial and C denotes the circle $|z - a| = R$, what is the value of $\int_C P(z) d\bar{z}$?

$\gamma : z = z(t), \alpha \leq t \leq \beta$. Consider $h(t) = \int_{\alpha}^t \frac{z'(t)}{z(t)-a} dt$. If $z'(t)$ is continuous, $h'(t) = \frac{z'(t)}{z(t)-a}$. The function $e^{-h(t)}(z(t) - a)$ has derivative

$$e^{-h(t)}(-h'(t)(z(t) - a) + z'(t)) = 0,$$

so $z(t) - a = Ce^{h(t)}$ for some $C \in \mathbb{C}^{\times}$. Since $z(\alpha) = z(\beta)$, we have $Ce^{h(\alpha)} = Ce^{h(\beta)} \rightarrow e^{h(\beta)} = 1 \rightarrow \beta = 2\pi in$ for some $n \in \mathbb{Z}$. Denote by $n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z-a}$.

Let $z : I \rightarrow \mathbb{C} - a$. For each $t \in I$, let Ω_t be the complement of the ray from 0 to $-z(t) + a$, which is well-defined since $z(t) \neq a$. Note $z(t) - a \in \Omega_t$. Since Ω_t is open and z is continuous, pick $\delta_t > 0$ such that $s \in V_{\delta_t}(t) \cap I \rightarrow z(s) \in \Omega_t$. $\{V_{\delta_t}\}_{t \in I}$ is an open cover of I .

Since I is compact, by Lebesgue Number lemma, pick a partition $0 = t_0 < t_1 < t_2 < \dots < t_n = 1$ of I such that $[t_k, t_{k+1}] \subseteq V_k$ for all $k \in [[n-1]]$. Let $z(V_k) \subseteq \Omega_k$. Then

$$J = \int_0^1 \frac{z'(t)}{z(t) - a} dt = \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} \frac{z'(t)}{z(t) - a} dt$$

$\log(z(t) - a)$ is analytic for $t \in [t_k, t_{k+1}]$ since $z(t) - a \in \Omega_k$. Hence,

$$J = \sum_{k=0}^{n-1} \log(z(t) - a) \Big|_{t=t_k}^{t_{k+1}}$$

Let $\log$ be analytic on regions Ω_1, Ω_2 . Suppose $z \in \Omega_1 \cap \Omega_2$. Then I claim $\log_1 z - \log_2 z = 2\pi il$ for some $l \in \mathbb{Z}$. Indeed, $e^{\log_1 z} = z = e^{\log_2 z} \rightarrow e^{\log_1(z) - \log_2(z)} = 1$, so $\log_1(z) - \log_2(z) = 2\pi il$, as desired. Since $t_0 = 0$, $t_n = 1$, and $z(0) = z(1)$,

we have $J = 2\pi im$ for some $m \in \mathbb{Z}$.

Let $\{t_i\}_{i \in [n]}$ be a partition of I with $t_0 = 0$ and $t_n = 1$. Let $\{\gamma_i\}, \{\sigma_i\}$ denote the associated subarcs and line segments from t_i to t_{i+1} . Denote by $\{t'_i\}_{i \in [n']}, \{\gamma'_i\}, \{\sigma'_i\}$ similarly and by $\{t''_i\}_{i \in [n'']}, \{\gamma''_i\}, \{\sigma''_i\}$ the common refinement. Consider the subarc γ_i , which is contained in a disc D_i . Suppose $i' = i$ and $j' = i + 1$. Since D_i is convex, we have that σ_i and σ''_k for $i'' \leq k < j''$ are contained in D_i . Since D_i doesn't contain a , the integral of $1/(z - a)$ over the loop $\sigma_i - (\sigma'_{i''} + \sigma'_{i''+1} + \dots + \sigma'_{j''-1})$ is zero. It follows that

$$\int_{\sigma_0 + \sigma_1 + \dots + \sigma_{n-1}} \frac{dz}{z - a} = \int_{\sigma''_0 + \sigma''_1 + \dots + \sigma''_{n''-1}} \frac{dz}{z - a},$$

so the definition is independent of the partition $\{t_i\}$.

If γ is piecewise differentiable, then $\gamma_k - \sigma_k$ is a (piecewise differentiable) loop contained in a disc D_k not containing a . The same reasoning above can be applied.

$$n(\gamma, 0) = 1, \text{ so } \int_{|z|=1} \frac{e^z}{z} dz = e^0 = 1.$$

$z = \rho(\cos \theta + i \sin \theta) \rightarrow dz = \rho(-\sin \theta + i \cos \theta)d\theta$. $|dz| = \rho d\theta$. Then

$$\begin{aligned} dz &= (-\sin \theta + i \cos \theta)|dz| = i(\cos \theta + i \sin \theta)|dz| \\ &= i(z/\rho)|dz| \rightarrow \\ |dz| &= -i \frac{\rho dz}{z} \end{aligned}$$

Then

$$\begin{aligned} \int_{|z|=\rho} \frac{-i\rho dz}{z(z-a)(\bar{z}-\bar{a})} &= \int_{|z|=\rho} \frac{-i\rho dz}{z(z-a)(\rho^2/z-\bar{a})} \\ &= -i\rho \int_{|z|=\rho} \frac{dz}{(z-a)(\rho^2-\bar{a}z)} = -\frac{i\rho}{\rho^2-|a|^2} \int_{|z|=\rho} \left(\frac{\bar{a}}{\rho^2-\bar{a}z} + \frac{1}{z-a} \right) dz \end{aligned}$$

If $a = 0$, the integral becomes $-\frac{i\rho}{\rho^2-|a|^2} 2\pi i = \frac{2\pi\rho}{\rho^2-|a|^2}$. If $|a| < \rho$, then $|\rho^2/\bar{a}| > \rho$, so the integral is $-\frac{i\rho}{\rho^2-|a|^2} 2\pi i = \frac{2\pi\rho}{\rho^2-|a|^2}$. If $|a| > \rho$, then the integral is $-\frac{i\rho}{\rho^2-|a|^2} \bar{a} \cdot 2\pi i = \frac{2\pi\bar{a}\rho}{\rho^2-|a|^2}$. Note that $a = 0$ and $0 < |a| < \rho$ can be combined into one case.

Suppose $\varphi(\zeta)$ is continuous on the arc γ . Then the function $F_n(z) = \int_{\gamma} \frac{\varphi(\zeta)d\zeta}{(\zeta-z)^n}$ is analytic in each of the regions determined by γ and its derivative is $F'_n(z) = F_{n+1}(z)$.

We first show that $F_1(z)$ is continuous. Let Ω be a connected component of

$\mathbb{C}^* - \gamma(I)$ and pick $z_0 \in \Omega$. We have for z sufficiently close to z_0 that

$$\begin{aligned} J &= |F_1(z) - F_1(z_0)| = \left| \int_{\gamma} \left(\frac{\varphi(\zeta)}{\zeta - z} - \frac{\varphi(\zeta)}{\zeta - z_0} \right) d\zeta \right| \\ &= \left| \int_{\gamma} \frac{\varphi(\zeta)(z - z_0)}{(\zeta - z)(\zeta - z_0)} d\zeta \right| \leq \int_{\gamma} \frac{|\varphi(\zeta)| |z - z_0|}{|\zeta - z| |\zeta - z_0|} |d\zeta| \end{aligned}$$

Let $\epsilon > 0$. Since $\gamma(I)$ is closed and $z_0 \notin \gamma(I)$, we have that $l = d(z_0, \gamma(I)) > 0$. Let $z \in B(z_0, l/2)$. Then

$$J \leq |z - z_0| \cdot (2/l) \cdot \int_{\gamma} \frac{|\varphi(\zeta)|}{|\zeta - z_0|} |d\zeta|$$

Denote the constant on the RHS by C . Let $\delta = \min(l/2, 1/(C+1)) > 0$. Then for $z \in B(z_0, \delta)$, we have $J \leq \epsilon$, so J is continuous.

$\varphi(\zeta)/(\zeta - z_0)$ is continuous on γ . Applying the above result, we have

$$\begin{aligned} F_1'(z_0) &= \lim_{z \rightarrow z_0} \frac{F_1(z) - F_1(z_0)}{z - z_0} = \lim_{z \rightarrow z_0} \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z)(\zeta - z_0)} d\zeta \\ &= \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^2} d\zeta = F_2(z_0). \end{aligned}$$

Assume the result for n . $\varphi(\zeta)/(\zeta - z_0)^n$ is continuous on γ . Then

$$\begin{aligned} F_n(z) - F_n(z_0) &= \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z)^{n-1}(\zeta - z_0)} - \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z_0)^n} + (z - z_0) \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z)^n(\zeta - z_0)} \\ \lim_{z \rightarrow z_0} \frac{F_n(z) - F_n(z_0)}{z - z_0} &= (n-1) \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z_0)^{n+1}} + \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z_0)^{n+1}} = nF_{n+1}(z_0). \end{aligned}$$

I claim that $G(z) = \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z)^n(\zeta - z_0)}$ is continuous. Let $\epsilon > 0$. Pick $\delta > 0$ such that $B(z_0, \delta) \subseteq \mathbb{C}^* - \gamma(I)$. Since $\gamma(I)$ and $\overline{B(z_0, \delta/2)}$ are compact, let $M = \max\{|\zeta - z| : \zeta \in \gamma(I), z \in \overline{B(z_0, \delta/2)}\}$. Then for $z \in B(z_0, \delta/2)$, we have

$$\begin{aligned} |G(z) - G(z_0)| &= \left| \int_{\gamma} \frac{\varphi d\zeta}{(\zeta - z)^n(\zeta - z_0)} - \frac{\varphi d\zeta}{(\zeta - z_0)^n(\zeta - z_0)} \right| \\ &= \left| \int_{\gamma} \frac{\varphi((\zeta - z_0)^n - (\zeta - z)^n)}{(\zeta - z_0)^n(\zeta - z)^n(\zeta - z_0)} d\zeta \right| \\ &\leq \left| \int_{\gamma} \frac{\varphi(z - z_0) \sum_{k=0}^{n-1} (\zeta - z_0)^k (\zeta - z)^{n-1-k}}{(\zeta - z_0)^n(\zeta - z)^n(\zeta - z_0)} d\zeta \right| \\ &\leq |z - z_0| \int_{\gamma} \frac{|\varphi| \sum_{k=0}^{n-1} |\zeta - z_0|^k M^{n-1-k}}{|\zeta - z_0|^{n+1} (\delta/2)^n} |d\zeta|. \end{aligned}$$

The constant on the right is independent of z , so G is indeed continuous at z_0 . This justifies the above calculation.

Liouville's theorem: A function which is analytic and bounded in the whole plane must reduce to a constant.

Proof: Let f be bounded by M . Let $z \in \mathbb{C}$ and denote by C the circle at the origin with radius $r > |z|$. We have

$$\begin{aligned} |f^{(n)}(z)| &= \left| n! \int_C \frac{f(\zeta) d\zeta}{(\zeta - z)^{n+1}} \right| \leq \int_C \frac{n! |f(\zeta)| |d\zeta|}{|\zeta - z|^{n+1}} \\ &\leq \int_C \frac{n! M |d\zeta|}{(r - |z|)^{n+1}} = \frac{2\pi n! r M}{(r - |z|)^{n+1}} \end{aligned}$$

For $n > 0$, we have $\lim_{r \rightarrow \infty} \frac{2\pi n! r M}{(r - |z|)^{n+1}} = 0$, so $f^{(n)}(z) = 0$, so f is constant.

Fundamental theorem of algebra. Let P be a polynomial of degree > 0 . Suppose FTSOC P has no roots. Pick a disc with radius r centered at the origin such that $|P(z)| \geq 1$ for $|z| \geq r$. Since $\overline{B}(0, r)$ is compact and $|P(z)|$ is continuous, let ϵ denote the minimum value of $|P(z)|$. Then $|P(z)| \geq \min(1, \epsilon) > 0$ for all $z \in \mathbb{C}$, so $f(z) = 1/P(z)$ is bounded entire. By Liouville's theorem, we have $f(z) = C = 1/P(z)$, so P is constant, absurd.

$\int_{|z|=1} \frac{e^z}{z^n} dz = \frac{2\pi i}{(n-1)!}$ by Cauchy for $n > 0$. For $n = 0$, we have $\int_{|z|=1} e^z dz = 0$ since $e^z = (e^z)'$.

$z^n(1 - z)^m$ is a polynomial and therefore the derivative of an analytic function. Hence, $\int_{|z|=2} z^n(1 - z)^m dz = 0$.

Claim: Let f be analytic on Ω and z_0 be a limit point of $Z(f)$. Then $z_0 \in Z(f^{(n)})$ for all $n \geq 0$.

Proof: Let $\{z_n\}$ be a sequence in $Z(f)$ converging to z_0 not equaling z_0 . Since f is continuous, we have $f(z_0) = 0$. Assume the result for k . Suppose FTSOC $f^{(k+1)}(z_0) \neq 0$. Then

$$f^{(k+1)}$$